

Project Design Phase-II Technology Stack (Architecture & Stack)

Date	25 June 2026
Team ID	
Project Name	An employee requests installation of licensed software through the Service Catalog
Maximum Marks	4 Marks

Technical Architecture:

The Deliverable shall include the architectural diagram as below and the information as per the table1 & table 2

Example: Software Installation Request System

Reference: <https://developer.ibm.com/patterns/ai-powered-backend-system-for-order-processing-during-pandemics/>

Project: Software Installation Request System (ServiceNow)

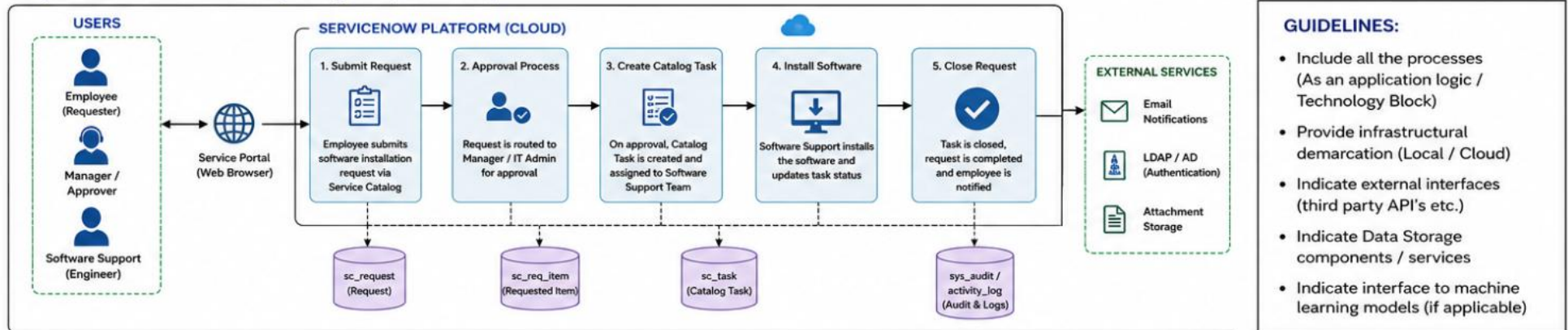


Table-1 : Components & Technologies:

S.No.	Component	Description	Technology
1.	 User Interface	Employees interact with the system through ServiceNow Service Portal to submit and track software installation requests.	ServiceNow Service Portal, Service Catalog
2.	 Application Logic-1	Validates request details, captures required variables and creates Service Catalog Request (REQ) and Requested Item (RITM).	ServiceNow Catalog Item, UI Policies, Business Rules
3.	 Application Logic-2	Routes requests for approval to Manager / IT Admin and handles approval workflow.	Workflow Editor, Flow Designer, Approval Policies
4.	 Application Logic-3	Creates Catalog Tasks (SCTASK), assigns to Software Support Team and updates task & request status based on progress.	Workflow Activities, Business Rules, Assignment Rules
5.	 Database	Stores all application data including Requests, Requested Items, Catalog Tasks, Users, and Variables.	ServiceNow Tables (sc_request, sc_req_item, sc_task, sys_user, item_option_new, etc.)
6.	 Cloud Database	ServiceNow cloud database for storing and managing application data.	ServiceNow Cloud Platform
7.	 File Storage	Stores attachments uploaded with software requests (e.g., licenses, justifications, documents).	ServiceNow Attachment Storage
8.	 External API-1	Sends email notifications for request submission, approvals, task assignment and completion.	ServiceNow Notification Engine (SMTP)
9.	 External API-2	Authenticates users and validates roles from corporate directory.	LDAP / Active Directory (Optional)
10.	 Machine Learning Model	Not applicable for current scope.	N/A
11.	 Infrastructure (Server / Cloud)	Cloud hosted ServiceNow platform for application deployment and execution.	ServiceNow SaaS Cloud

Activate Windows

Table-2: Application Characteristics:

S.No.	Characteristics	Description	Technology
1.	 Open-Source Frameworks	Uses ServiceNow platform which is built on modern open technologies with JavaScript for customizations.	JavaScript, ServiceNow Platform
2.	 Security Implementations	Implements role-based access, ACLs, approvals, audit logs and data encryption in transit.	ACLs, User Roles, HTTPS, ServiceNow Security
3.	 Scalable Architecture	Cloud-based architecture supports multi-tenant environment, multiple catalogs, workflows and large number of users.	ServiceNow SaaS Architecture
4.	 Availability	High availability provided by ServiceNow cloud infrastructure with redundancy and backups.	ServiceNow Cloud Infrastructure
5.	 Performance	Automated workflows, assignments and catalog tasks improve processing speed and reduce manual effort.	Workflow Editor, Flow Designer, Business Rules

References:

<https://c4model.com/>

<https://developer.ibm.com/patterns/online-order-processing-system-during-pandemic/>

<https://www.ibm.com/cloud/architecture>

<https://aws.amazon.com/architecture>

<https://medium.com/the-internal-startup/how-to-draw-useful-technical-architecture-diagrams-2d20c9fda90d>

